THE WORLD OF CHARLESTON SOUTHERN UNIVERSITY

A Virtual Reality Tour of Campus

Thomas Finch

Advisor: Yu-Ju (Jospeh) Lin, PhD

Charleston Southern University
Computer Science Senior Project

Introduction

Purpose:

The purpose of this project is to use the skills learned from multiple courses taken at Charleston Southern University and apply this knowledge to a real world application. With guidance and recommendations from Dr. Lin, I opted to create an application for Android based smart phones. I felt this would give me the opportunity to better understand how smart phone application are created and how they can be useful to others.

The application I chose to make, with the advisement of Dr. Lin, was a Virtual Reality Tour of the CSU Campus using Google Cardboard. This application was created for perspective students, new students, current students, alumni, and anyone else interested in learning the more about Charleston Southern University. Google Cardboard is inexpensive and can be used with most smartphones whether it is Android or iOS based. For perspective students, new students, and current students, the application will help them learn the location of various building and facilities on campus, help them recognize the buildings and facilities, and also show them how to navigate around campus without the fear of getting lost. For alumni, this app offers them a chance to visit the campus and see how the university has grown and changed since the last time they were able to visit the campus. For other people interested in learning more about Charleston Southern University, this app will showcase the beauty of the campus and also show them how it is continuing to grow.

The one thing I like most about this application is its expandability. As the campus grows and the appearance of buildings changed, the application can easily be updated to reflect these changes. In addition to this, audio can also be added, giving the user detailed information about the scene being viewed. On a final note, the platform used to create this application is capable of creating the app for Android and iOS phones.

Table of Contents

Required Materials:

For this project, I needed to use the following items:

- Google Cardboard – this can be purchased online
- Samsung Galaxy 5 – any Android phone using version 4.1 or higher
- Android Studio – one of the platforms recommended by Google Cardboard
- Unity 3D – one of the platforms recommended by Google Cardboard
- 360° Camera – used to obtain photos, I used an I-Phone
- Visual Studio 2013 – used to write/edit any required scripts

Development:

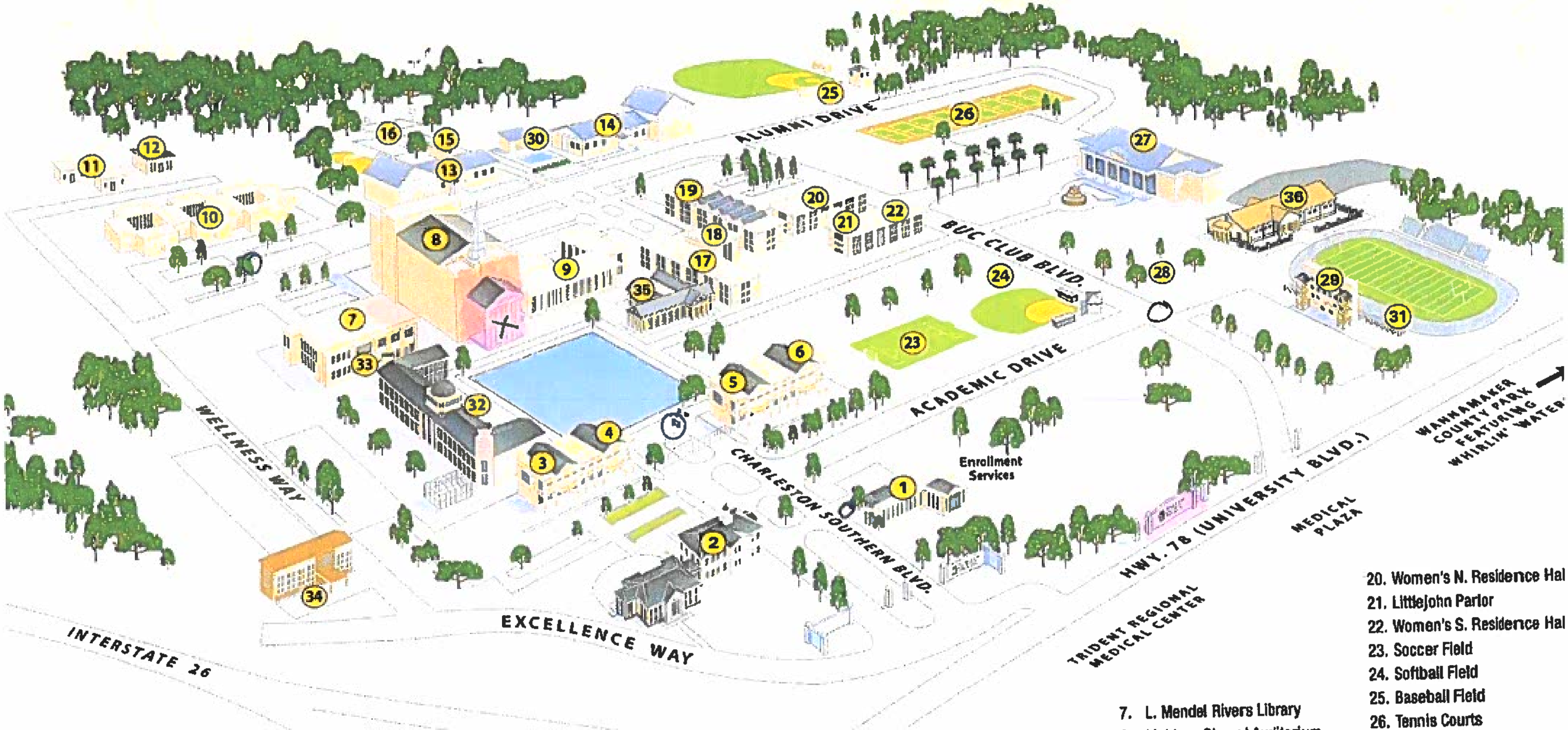
Step 1 in developing this app was to research Google Cardboard. The website for Google Cardboard contains a site for assisting developers in getting started with the app. This site also contains the Google VR SDK for Unity 3D, Android, and iOS apps. Using the provided instructions, I created the demo app to become more familiar with how Google Cardboard interacts with Android Studio and Unity 3D. This allowed me to become familiar with both platforms. After reviewing Android Studio and Unity 3D, I chose to use Unity 3D because I felt it was more user friendly.

Step 2 in the development process was to determine how to obtain the photos of the campus. For this step, I knew I would need 360° pictures. To accomplish this, I purchased a Samsung Galaxy 5. The Galaxy 5 did not have an application to take 360° pictures, so I downloaded the Google Camera app, I began collecting 360° pictures of the campus. One issue I ran into during this process was the quality of the pictures. The quality of the 360° pictures were not the level I had expected, but it did allow me to begin developing the application. I later replaced the original 360° pictures with ones I took with an I-Phone 6 using the Google Street application. Using the campus map provided on the CSU website, I organized the photos to form the overall map of the campus.

Step 3 in the development process was formatting the pictures and creating a working demo to view them. The first major obstacle I faced in this process was creating a photosphere I could use in the application. To achieve this, I used a C# script that creates an Inverted Sphere and added the 360° pictures to create scenes. This allowed me to position the camera on the inside of the photosphere. To adjust for the picture being in reverse I changed the settings of the x-axis in the scale to -1 on the Inverted Sphere. At this point, I was able to view the campus scenes using Google Cardboard, but I was not able to navigate around campus.

Step 4 in the development process was to create a way to navigate. Accomplishing the navigation was a three-step process. The first step was to create clickable buttons that would allow the user to choose the direction to move. To accomplish this, I created buttons by inserting text into the scene using the Text Mesh and then added Box Colliders to the text. The box colliders allowed me to define the area around the button that would be clickable. In addition to adding buttons, I color coded the button to identify if the area would be Academic, Administrative, Athletic, or Student Activities. Once I had the buttons created I added them to the scenes. Having positioned the buttons in the proper locations, I need to find a way to interact with them. The second step partially accomplished this by using the Cardboard Reticle. The Cardboard Reticle is a pointer that shows the user where they are looking. In this case, I choose a circle that would increase in size when it was over a button. At this point in the process, I was able to view the scene and point at the button using the reticle, but I still could not interact with the button. The third and final step in this process was to create a script using C# that would connect the buttons with the proper scene. This process was repeated for all the buttons used to navigate the campus except the menu button and the buttons used on the opening scene to jump to the different areas on campus. For the main button, I created a canvas that would allow me to copy and paste it to each scene. This allowed me to ensure it was in the same location for each scene. I also used a canvas on the opening scene to ensure there was proper spacing between the buttons.

Step 5 in the development process was to design the map. With the aid of the campus map provided on the CSU website, I added the proper buttons to each scene. Positioning the buttons did pose some difficulty because the buttons are not part of the photosphere. Once this was accomplished, the only thing left was to begin testing.



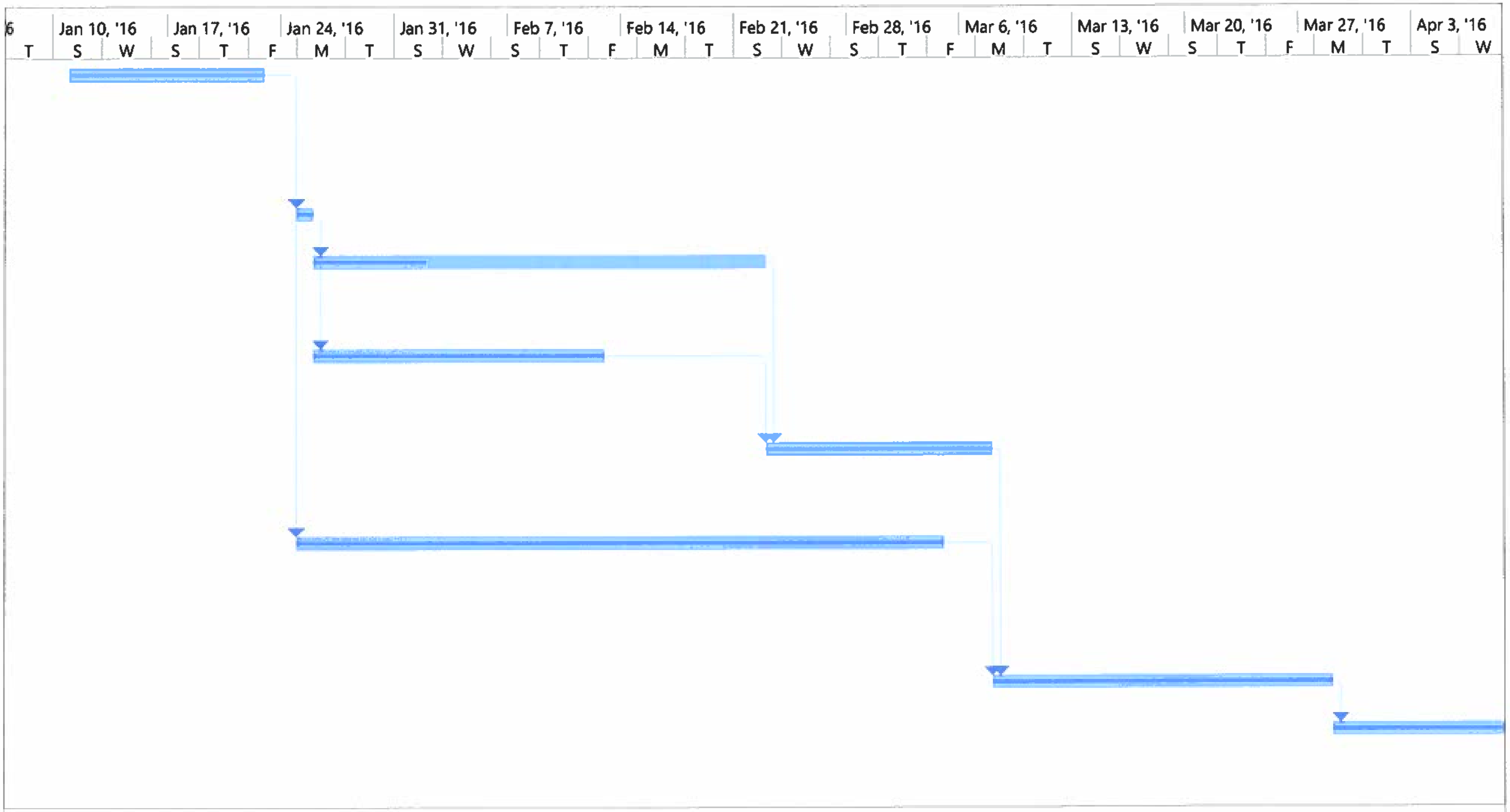
CHARLESTON SOUTHERN UNIVERSITY

1. Hunter Reception Center
   - Admissions
   - Registrar's Office
   - Financial Aid
   - Student Accounts
   - Student Employment
   - Veterans Services
2. College of Nursing and Allied Health
3. Wingo Hall
4. Norris Hall
5. Jones Hall
6. Ashby Hall
7. L. Mendel Rivers Library
8. Lightsey Chapel Auditorium Lightsey Music Building
9. Strom Thurmond Center
10. Quads 1, 2, 3 (Residence Halls)
11. Physical Plant
12. Art Building
13. Whittington Hall
14. Brewer Center
15. Communications Center
16. University Park
17. Russell West Residence Hall
18. Learning Center
19. Russell East Residence Hall
20. Women's N. Residence Hal
21. Littlejohn Parlor
22. Women's S. Residence Hal
23. Soccer Field
24. Softball Field
25. Baseball Field
26. Tennis Courts
27. Field House
28. Disc Golf Course
29. Whitfield Stadium Center
30. Patsy Morley Pool
31. Bagwell-Settle Track and CSU Stadium
32. Science Building
33. Java City Café
34. Wingate by Wyndham *(located on the CSU campu*
35. Whitfield Center for Christian Leadership
36. Athletic Center

ID		Task Mode	Task Name	Duration	Start	Finish	Predecessors	Critical	n 3, M
1	✓		Research the different types of Apps for 360 pictures	10 days	Mon 1/11/16	Fri 1/22/16		No	
2	✓		Download the App	1 day	Mon 1/25/16	Mon 1/25/16	1	No	
3			Take photos of the Campus	20 days	Tue 1/26/16	Mon 2/22/16	2	No	
4	✓		Obtain Google Cardboard	14 days	Tue 1/26/16	Fri 2/12/16	2	No	
5	✓		Organize the campus photos	10 days	Tue 2/23/16	Mon 3/7/16	3,4	No	
6	✓		Research the proper format to display the photos	30 days	Mon 1/25/16	Fri 3/4/16	1	No	
7	✓		Create a working model	15 days	Tue 3/8/16	Mon 3/28/16	5,6	No	
8	✓		Test and work out any mishaps	15 days	Tue 3/29/16	Mon 4/18/16	7	No	

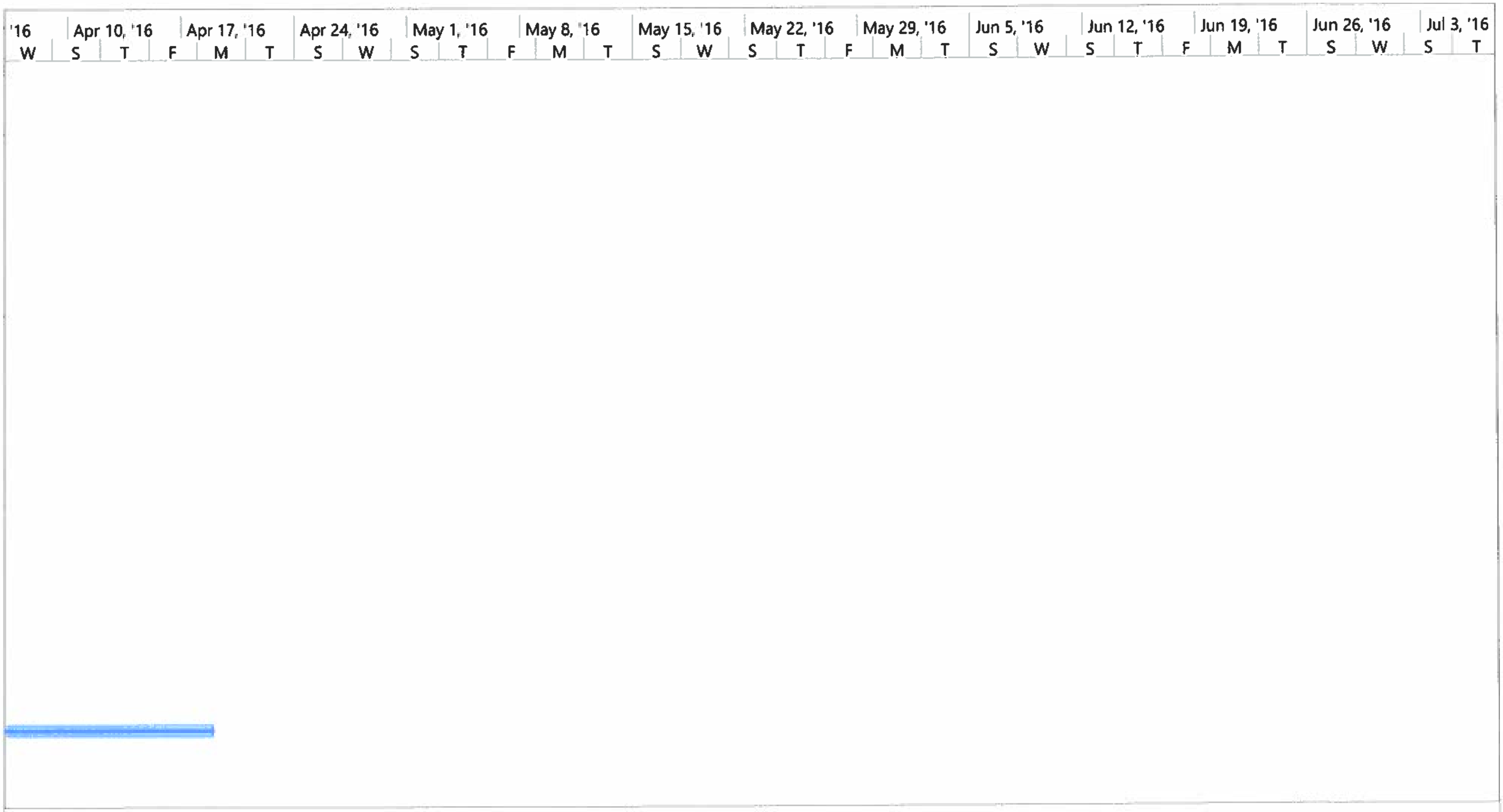
Project: senior project timeline
Date: Sat 11/12/16

Task	Inactive Summary	External Tasks
Split	Manual Task	External Milestone
Milestone	Duration-only	Deadline
Summary	Manual Summary Rollup	Critical
Project Summary	Manual Summary	Critical Split
Inactive Task	Start-only	Progress
Inactive Milestone	Finish-only	Manual Progress



Project: senior project timeline
Date: Sat 11/12/16

Task	Inactive Summary	External Tasks
Split	Manual Task	External Milestone
Milestone	Duration-only	Deadline
Summary	Manual Summary Rollup	Critical
Project Summary	Manual Summary	Critical Split
Inactive Task	Start-only	Progress
Inactive Milestone	Finish-only	Manual Progress



Project: senior project timeline
Date: Sat 11/12/16

Task	Inactive Summary	External Tasks
Split	Manual Task	External Milestone
Milestone	Duration-only	Deadline
Summary	Manual Summary Rollup	Critical
Project Summary	Manual Summary	Critical Split
Inactive Task	Start-only	Progress
Inactive Milestone	Finish-only	Manual Progress

The CSU Virtual Tour App

Which is the better way to interact with this app, Google Cardboard or the touch screen

Thomas Finch

Computer Science Department

Charleston Southern University

Charleston, SC

Abstract

Google Cardboard is one of the newest technologies to hit the market. This allows the user to interact with apps to play games, take tours, and view photos in a virtual environment. Building upon this form of interaction, the Charleston Southern University Virtual Tour of Campus app allows users to take a tour of this beautiful campus. In this study, it will be determined if there is a significant difference in the performance of this app with first time users utilizing Google Cardboard and the touch screen on android based phones. This study will analyze both the time required to navigate through the campus and the number of errors committed using both forms of interaction. In addition to this, the participants will provide valuable feedback on their experience and the preferences when interacting with the app. From this data, it can be determined that both methods

of interaction for first time users perform similarly, but Google Cardboard provides a more enjoyable, fuller experience.

I. Introduction

As new technology is being developed at an increasing rate, one way Charleston Southern University can set itself apart from other universities by offering a unique format to showcase the campus. For this study, the new technology I am referring is an interactive virtual map of Charleston Southern University designed to be used on Android phones. The purpose of this app is to create a virtual tour of campus the user can interact with either by utilizing Google Cardboard or the touch screen on their android phone. It is important for institutes of higher learning to keep pace with technology and showcase not only how beautiful the campus is, but to demonstrate some of the technologies the students are learning to utilize. A Virtual Tour of Campus using Google Cardboard accomplishes both of these things. This app was created using Unity app developer, which allows you to create an android application in 3D for virtual reality. Participants in this study were allowed to explore the app utilizing both the touch screen features of an android phone and Google Cardboard. Once the participant felt comfortable with the app, they were asked to find a location on campus using both methods for a time analysis. The participants first utilized the Google Card Board and then the touch screen features while being recorded. After completing the app testing, the participants were asked to complete a questionnaire about their experience. The purpose of this study is to determine:

- Is it significantly faster using one method verse the other?
- Did the users prefer to interact with the map application utilizing the touch screen method or Google Cardboard method?
- Were the navigation tools, which include a menu that allows the user to jump to various locations on campus and color coded links to denote the type of areas available, are helpful and easy to understand?
- Does the app provide a proper depiction of the campus?
- Was the app is helpful in learning the layout of the campus?
- Was the app difficult to use?
- Does the app create an enjoyable experience?

The results from this study will help to design an app that will fulfill the user's expectations and lead to further development of this app to include new features and expanded upon the overall experience.

II. Related Work:

For the development of this app, research was performed through studying the information provided from the Google Cardboard Developer web pages. These web pages provided the basic information to start learning how to create objects, manipulated them., and the required SDK files for both Android Studio and Unity 3D. The Unit 3D developer web pages provided insight on how to utilize the features provided within the Unity program. In addition to the research on the app development, various apps were researched and tested to obtain the proper photospheres used to create the images used in the app. This consisted of research apps compatible for both android based phones and IPhones. The IPhone presented the clearest photospheres using the Google Street app. Research was also conducted on recording apps to be used during the testing phase. This provided information such as the amount of time participants needed to complete the assigned tasks and a record of any errors committed.

III. System Description:

This app was created using a Windows based laptop computer. The program used to design the app was, as stated above, Unity 3D with the Google Cardboard SDK. This app is designed for Android based smart phones with an operating system of Android 4.1 Jellybean. The Google Cardboard SDK allows for the screen to be duplicated for each eye and is designed to be used in landscape view (sideways) with Google Cardboard, but can also be used with the touch screen options provided on smart phones. Google Cardboard is an open cardboard box with a flap on the back where the user will place there android phone. The front side of the box is designed to fit flush against the users face. On the inside, the box contains two lenses. The button on the outside of the Google Cardboard varies between design types. On the Google Cardboard used for this prototype, the button is located on the top, right side. When this is pressed, the lower half on the button touches the screen to make the selection. On other versions of Google Cardboard, a magnet is placed on the left side and a selection can be made by sliding this in a downward motion. When the app is running, the box blocks out the surrounding light and the lenses create a sense of depth in the screen. The images displayed on the screen are photospheres created using a photosphere camera app. An alternative option to using a photosphere app is a 360° camera. The images are then wrapped around a sphere, turned inside out, and then reverse. On the screen main screen, the user is presented with a menu that allows instant navigation to any of the four primary locations on campus. To navigate directly to the Academic area of the campus, the user would select the

Academics, which is written in blue. All the links that lead to an academic location are in blue. Athletics is in gold and all links that are in gold will take you to another portion of the athletic area on campus. Student Housing is coded in brown. All links that are in brown will either take you to Student Housing or a student recreation area. The links that are in green are used to take you to Business offices on campus. An easy way to remember this is the school colors are blue and gold. Academics are more important than athletics, so, blue is for academic and gold is for athletics. Brown is used for the Student Housing and recreation because brown in an earthy color. And finally, since most business transaction require money, the business offices are green. Each screen also contains a link to the areas surrounding the current location. These links are color coded to correspond to the four primary locations on the map.

IV. Prototype trials and evaluations

A. Participants:

All of the participants in this evaluation were professionals in their field over the age of 20. There were a total of 5 participant. The ages and sex of the participants are broken down as follows:

- 1 male between the ages of 20 to 29

- 1 male between the ages of 30 to 39
- 1 female over the age of 40
- 2 males over the age of 40

The professions of the male participants are: 2 telecommunications technicians, 1 technician, and 1 Electrical Engineer. The profession of the female participant is a baker/cake decorator. The participants each have different levels of experience using Android phones.

B. Experimental Design:

This experiment is design to test how well the user can utilize Google Cardboard and compare those results to how well they are able to accomplish the same task using the touch screen features on the android phone. The experiment will consist of the participant using the app to navigate through the campus using both Google Cardboard and using the touch screen. After completing a familiarization period using the app with both forms of interaction, the participants will be given one opportunity to complete each task. These tasks will be recorded by using a screen recorder app that will provide a record of the amount of time taken to complete each task along with any errors committed. The participants will be allowed to choose any path they feel is best to reach the final destination. Upon completion of both tasks, the participants will be given a questionnaire to complete about their experience.

C. Apparatus:

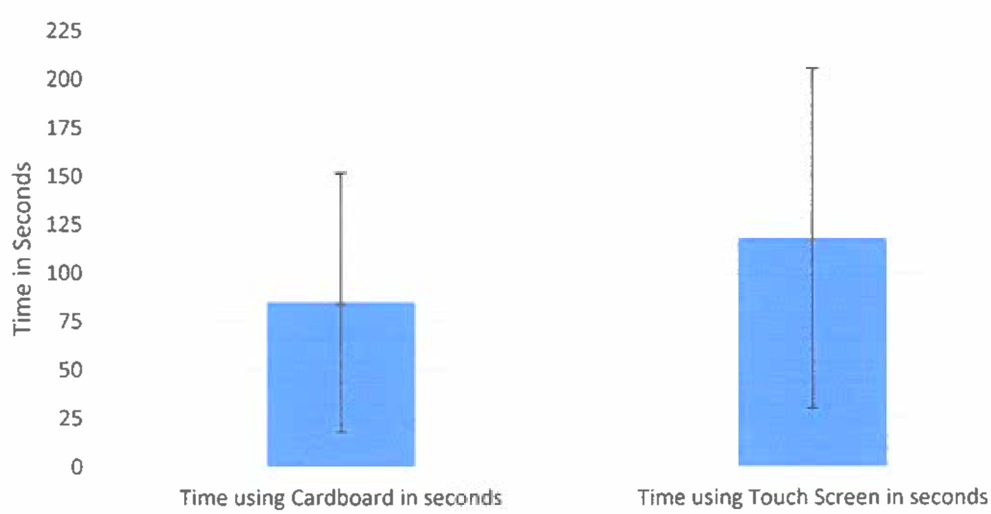
The condition being tested is whether Google Cardboard or the touch screen is easy to use to navigate the campus with this app. After completing the consent form, the presenter will provide a paper copy of the campus map for the participant to review. The participants will then be asked to watch two video depicting the onscreen use of the app and the navigation options. The first video will first demonstrate the 360° capabilities of the virtual environment the app produces. Then, the video will demonstrate how the main screen menu works by linking to each of the four campus areas. The second video will demonstrate how to navigate through the campus using the scene links. The presenter will then demonstrate how the touch screen works to the participant. Next, the presenter will explain how the Google Cardboard works and demonstrate the controls on the box. The participant will then be given 5 minutes to explore the controls for each technique and to explore the app. During this time, the participant will be encouraged to ask any questions they may have about the app and the different interfaces. The task given to the participants is designed to allow the participant to determine their own route to reach the destination. For this study, the participants were asked to navigate from the start screen to the Lightsy Chapel. This location was chosen because if can be seen from multiple locations around campus and has multiple avenues to reach it. The purpose of this experiment is to determine if Google Cardboard or touch screen is the better interface when using this app. In addition to this the participant will be asked if the color coding is appropriate, if the campus layout is properly depicted, and if this app is a useful tool to learn the campus. During the testing phase, the presenter will start the recording app and then start the app. Once the app is running, the presenter will either place the Android phone inside Google Cardboard or hand the phone to the participant, depending upon which task is being performed. When the participant has reached the destination, the presenter will stop the recording and save the file for future analysis.

V. Results:

A. Time Trials

By utilizing a screen records, the time required to complete the task and the number of errors committed by the user were recorded for analysis. This data provided the information to perform a quantitative analysis. During the timed testing, the average difference between using Google Cardboard and the touch screen was 40.2 seconds with a standard deviation of 69.4709 seconds and the . This result is skewed due to the fact that one participant completed the Google Cardboard 164 seconds quicker than the touch screen, while the largest difference for the other participants was only 18 seconds. The average time required to complete the task using Google Cardboard was 85 seconds with a standard deviation of 53.9212 seconds and the average for the touch screen was 118 seconds with a standard deviation of 70.7955 seconds. Comparing the results from these two task using a 95% confidence interval, in the Chart 1 below, there does not appear to be a significant difference in the results.

Chart 1: Time Trials with 95% confidence



This result can be verified by performing a T-Test. The results from the T-Test two-tail showed the T value is 0.431, which is greater than 0.05. This is show in the Chart 2 below.

Chart 2 t-Test: Two-Sample Assuming Equal Variances		
	Time using Cardboard in seconds	*Time using Touch Screen in seconds*
Mean	85	118
Variance	2907.5	5012
Observations	5	5
Pooled Variance	3959.75	
Hypothesized Mean Difference	0	
df	8	
t Stat	-0.829182372	
P(T<=t) one-tail	0.215522377	
t Critical one-tail	1.859548038	
P(T<=t) two-tail	0.431044754	
t Critical two-tail	2.306004135	

For this study, an error consisted of making an incorrect choice while navigating to the goal. As shown in Chart 3, more of the participants created an error during the Google Cardboard portion of the testing than during the touch screen, but the average number of errors for each task is the same.

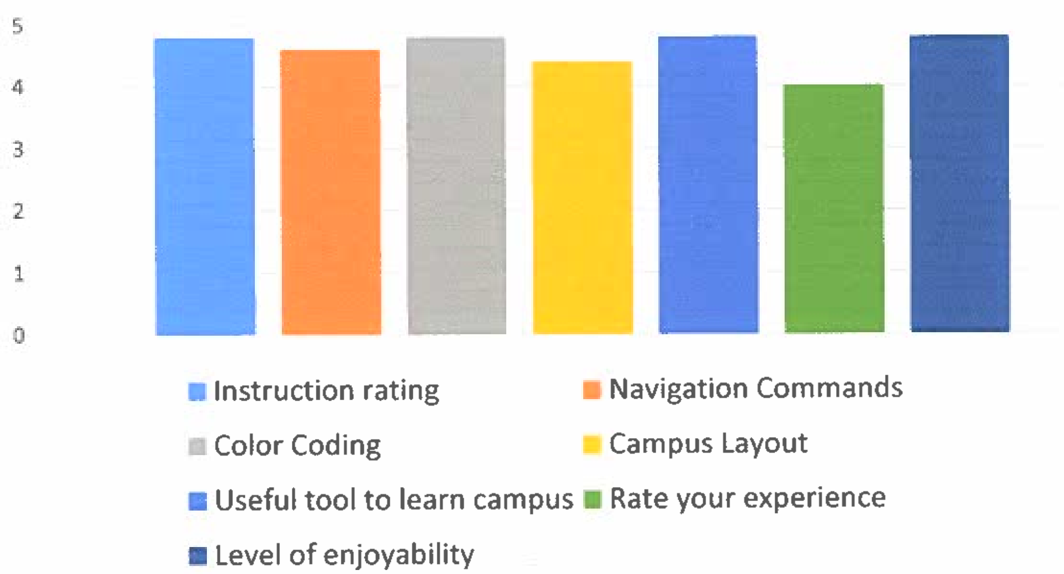
Chart 3: Participant Errors			
Participants	# of Errors committed with Google Cardboard	# of Errors committed with the Touch screen	Error Difference GC - touch
User 1	0	3	3
User 2	3	3	0
User 3	2	0	2
User 4	0	0	0
User 5	1	0	1
Average	1.2000	1.2000	1.2000
Median	1.0000	0.0000	1.0000
Standard Deviation	1.3038	1.6432	1.3038

The results from the time trials and errors show the is no real difference between navigating through the app utilizing Google Cardboard or the touch screen n the android phone.

B. Questionnaire

Upon completing the time trials, the participants were asked to complete a short questionnaire about their experience using the app. The questionnaire provides the qualitative data for this study using the Likert Scaling between 1 and 5, with 5 being the highest rating. As stated above, the study consisted of 5 participants, one female and four males, all of whom are professionals in their given field. None of the participants had prior experience using Google Cardboard, but all of them did own touch screen phones. The participants found the experience using the app to be very enjoyable (4.8 out of 5). They also found the campus layout on the app was very accurate (4.4 out of 5) and felt this would be a very useful tool in learning the campus layout (4.8 out of 5). The color coding used was easy to understand and very helpful during the demonstration (4.8 out of 5). When asked to rate their experience, one participant did cite feeling slightly dizzy after using both Google Cardboard and the touch screen. Even with this lower rating, the participants found their experience using the app easy (4.0 out of 5). The final question on the questionnaire asked which experience they preferred and all of them preferred using the Google Cardboard with this app. See Chart 4.

Chart 4: Questionnaire Averages



VI. Discussion:

The results from the timed trials did not support my hypothesis that one method would be significantly faster than the other. I felt Google Cardboard would be the faster and easier method to navigate through the campus. I think this result is due to multiple factors. The first one being that none of the participants had any prior experiences using Google Cardboard. This lack of experience could have slowed their navigation process down. Secondly, the participants had very limited to no prior knowledge of the campus layout. This lead to some confusion during the navigation process. The final factor was the size of the group involved in the study. I believe had the group been considerably larger, at least 50 participants, there would have been more significant separation between the time trials. Across the range of ages in the study, I app performed how I had anticipated. All the participants enjoyed using the app and felt the navigation commands were easy to understand. One participant did comment that Google Cardboard did offer a more complete experience, but point of view between scenes did make it difficult to orient the direction you were facing. Correcting this issue would be a big plus. I believe there are two different ways this could be corrected. The easy way and most likely the most time consuming would be to create additional scenes for each scene that were oriented to the direction the user is facing from the previous screen. This would greatly increase the size of the app, but I do not think it would slow it down. The second way, which I believe would be more efficient and properly fix to orientation, would be to access the gyroscope in the phone to determine how the next scene would be positioned in relation to a compass. This would require more research into both Google Cardboard and Unity to see if it is even possible.

VII. Conclusion:

From this study, it can be determined that with no experience using Google Cardboard, the average user does not see a significant time savings or an increase in the number of errors committed between using Google Cardboard and the touch screen. This does not mean that the study was a failure, if anything it shows the app is well designed. The results from the questionnaire clearly show the participants enjoyed the experience of using the app and felt that it would be extremely useful in learning the campus layout. It also show the choice of colors used for the different links were easy to understand and appropriate. With more research in the orientation aspects of the scenes, this app can be an asset to potential and new students, visitors, and alumni wanting to learn their way around campus. In addition to helping users learn their way around campus, this app could also be used as a marketing tool to showcase both the beauty of the campus and the technology being utilized by its students. One of the great things about this app is it can be expanded to include more locations, the interior of the buildings, and even audio files to describe the history of the campus and other fun facts about the school.

References:

A new dimension - Designing for Google Cardboard - VR design guidelines. (n.d.). Retrieved April 23, 2016, from https://www.google.com/design/spec-vr/designing-for-google-cardboard/a-new-dimension.html#

A new dimension - Designing for Google Cardboard - VR design guidelines. (n.d.). Retrieved April 23, 2016, from https://www.google.com/design/spec-vr/designing-for-google-cardboard/a-new-dimension.html#

How to Start Your Game Development Community tutorial. (n.d.). Retrieved April 23, 2016, from https://unity3d.com/learn/tutorials/modules/beginner/your-first-game/how-to-start-your-game-development

Setup - Interactive patterns - VR design guidelines. (n.d.). Retrieved April 23, 2016, from https://www.google.com/design/spec-vr/interactive-patterns/setup.html#

CHARLESTON SOUTHERN UNIVERSITY

CONSENT TO PARTICIPATE IN RESEARCH

Title of Research: Charleston Southern University Virtual Reality Campus Tour

Funding Agency/Sponsor: none

What is the purpose of the research?
The purpose of this study is to determine if this application is easier to use with Google Cardboard or touch screen.

For how long am I expected to be in this study and how much of my time is required? This will require between 30 to 60 minutes to complete

Where is this study to take place?
In a natural environment

How many participants will be in this study?
This study will consist of 5 to 10 participants

What is my involvement for participating in this study?
You will be asked to complete this Consent Form, conduct a 5 minute familerization period with the application, participate in a recorded, timed exercise navigating to a preset location using Google Cardboard and using the touch screen, and complete a survey about your experience.

What are the risks and/or potential discomforts of participating in this study?
There is a possibility the participant ma become dizzy while utilizing the application.

What are the anticipated benefits of participating in this study?
This can possibly lead to the development of an application for potential students, current students, alumni, and visitors of Charleston Southern University to experience a virtual reality tour of campus to learn how to navigate the campus while at the same time showcasing the beauty of Charleston Southern's campus.

Will I be compensated for participating in this study?
Participants will receive a heartfelt "Thank You" and a smile.

What is an alternative procedure(s) that I can choose instead of participating in this study?
Participants are welcome to view demonstration videos created during development

How will my confidentiality be protected?

Only the information provided by the participant will be recorded. This information will not be shared with any outside parties and will be destroyed upn completion of the research.

Is my participation voluntary?
Participation is 100% voluntary

Can I stop participating at any time during this research without penalty?
You may stop participating at any time you feel uncomfortable or you no longer wish to continue with the research.

How can I withdraw from this study if I choose not to participate any longer?
To withdraw from the study, please inform the presentor/investigator.

Will I receive a copy of this informed consent document?
Participants will receive a photocopy of the consent form.

Who should I contact if I have questions regarding this study?
Thomas Finch, Student, (843)834-8228, TRFinch@CSUStudent.net

Have you read and fully understood the informed consent above? Yes No

Your name and signature below indicates you have read the information above, you have freely decided to participate in this research, and you certify that you are at least 18 years old.

Participant Name (please print): ______________________________

Participant Signature: ______________________________ **Date:** ____________

Investigator Signature: ______________________________ **Date:** ____________

Faculty Sponsor Signature: ______________________________ **Date:** ____________

Thomas R. Finch
Charleston Southern University Campus Tour

Questionnaire:

Participant Information

Name (optional): ______________________________

Gender: ☐ Female ☐ Male

Age: ☐ 20 – 29 ☐ 30 – 39 ☐ 40 and above

Occupation: ______________________________

For the following questions, please select between 1 – 5, with 1 being the lowest rating and 5 being the highest:

Were the instructions given prior to the testing adequate?

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

How would you rate the navigator commands? 1 being difficult, 5 being simple

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Do you find the color coding of the different campus areas helpful?

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Do you feel the design was a proper depiction of the campus layout?

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Do you feel this is a useful tool for people wanting to learn about the layout of the campus? ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

How would you rate your experience using this application? 1 being difficult, 5 being easy? ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Did you find using this application enjoyable? 1 being difficult, 5 being extremely enjoyable ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Do you have any prior experience using Google Cardboard? ☐ Yes ☐ No

Do you prefer using the application with or without the Google Cardboard?

____

Additional comments: ______________________________

____

____

```
using UnityEngine;
using UnityEngine.SceneManagement;
using System.Collections;

public class ButtonController : MonoBehaviour {

    // This is to change the scene to the Entrance when the Entrance button is activated
    void Start()
    { }
    // Update is called once per frame
    void Update ()
    { }
    public void StartGame ()
    {
        SceneManager.LoadScene("csu");
    }
    //3d text menu buttons found at the bottom of the scene to navigate
    public void mainButton(string mainButton)
    {
        SceneManager.LoadScene("csu");
    }
    public void academicsButton(string academics)
    {
        SceneManager.LoadScene("academics");
    }
    public void athleticsButton(string academicBuc)
    {
        SceneManager.LoadScene("academicBuc");
    }
    public void adminButton(string huunter)
    {
        SceneManager.LoadScene("hunter");
    }
    public void dormButton(string quads)
    {
        SceneManager.LoadScene("quads");
    }
    public void csuButton(string csu)
    {
        SceneManager.LoadScene("csu");
    }
    //3d text scene buttons to navigate between locations
    public void entranceButton (string entrance)
    {
        SceneManager.LoadScene("entrance");
    }
    public void entranceHunterButton(string entranceHunter)
    {
        SceneManager.LoadScene("entranceHunter");
    }
    public void hunterButton (string hunter)
    {
        SceneManager.LoadScene("hunter");
    }
    public void hunter2Button(string hunter2)
    {
        SceneManager.LoadScene("hunter2");
    }
    public void entranceAcademicButton(string entranceAcademic)
    {
        SceneManager.LoadScene("entranceAcademic");
    }
    public void bucHwy78Button(string bucHwy78)
    {
        SceneManager.LoadScene("bucHwy78");
    }
```

```
    public void dpWingoButton(string dpWingo)
    {
        SceneManager.LoadScene("dpWingo");
    }
    public void nursingButton(string nursing)
    {
        SceneManager.LoadScene("nursing");
    }
    public void pondButton(string pond)
    {
        SceneManager.LoadScene("pond");
    }
    public void jonesButton(string jones)
    {
        SceneManager.LoadScene("jones");
    }
    public void ashbyButton(string ashby)
    {
        SceneManager.LoadScene("ashby");
    }
    public void christian1Button(string christain1)
    {
        SceneManager.LoadScene("christian1");
    }
    public void christianButton(string christian)
    {
        SceneManager.LoadScene("christian");
    }
    public void pond2Button(string pond2)
    {
        SceneManager.LoadScene("pond2");
    }
    public void cafeteriaButton(string cafeteria)
    {
        SceneManager.LoadScene("cafeteria");
    }
    public void stromThurmanButton(string stromThurman)
    {
        SceneManager.LoadScene("stromThurman");
    }
    public void strom2Button(string strom2)
    {
        SceneManager.LoadScene("strom2");
    }
    public void russellWest1Button(string russellWest1)
    {
        SceneManager.LoadScene("russellWest1");
    }
    public void chapelButton(string chapel)
    {
        SceneManager.LoadScene("chapel");
    }
    public void pond3Button(string pond3)
    {
        SceneManager.LoadScene("pond3");
    }
    public void libraryButton(string library)
    {
        SceneManager.LoadScene("library");
    }
    public void library2Button(string library2)
    {
        SceneManager.LoadScene("library2");
    }
    public void wellnessLibraryButton(string wellnessLibrary)
    {
```

```
        SceneManager.LoadScene("wellnessLibrary");
    }
    public void scienceButton(string science)
    {
        SceneManager.LoadScene("science");
    }
    public void pond4Button(string pond4)
    {
        SceneManager.LoadScene("pond4");
    }
    public void hcWingoButton(string hcWingo)
    {
        SceneManager.LoadScene("hcWingo");
    }
    public void wellnessWingoButton(string wellnessWingo)
    {
        SceneManager.LoadScene("wellnessWingo");
    }
    public void norrisButton(string norris)
    {
        SceneManager.LoadScene("norris");
    }
    public void academicSoccerButton(string academicSoccer)
    {
        SceneManager.LoadScene("academicSoccer");
    }
    public void soccerButton(string soccer)
    {
        SceneManager.LoadScene("soccer");
    }
/*
    public void academicSoftballButton(string academicSoftball)
    {
        SceneManager.LoadScene("academicSoftball");
    }

    public void softballButton(string softball)
    {
        SceneManager.LoadScene("softball");
    }
*/
    public void academicBucButton(string academicBuc)
    {
        SceneManager.LoadScene("academicBuc");
    }
    public void stadiumButton(string stadium)
    {
        SceneManager.LoadScene("stadium");
    }
    public void trackButton(string track)
    {
        SceneManager.LoadScene("track");
    }
    public void athleticCenterButton(string athleticCenter)
    {
        SceneManager.LoadScene("athleticCenter");
    }
    public void bucFieldHouseButton(string bucFieldHouse)
    {
        SceneManager.LoadScene("bucFieldHouse");
    }
    public void fieldHouseButton(string fieldHouse)
    {
        SceneManager.LoadScene("fieldHouse");
    }
    public void bucWomenButton(string bucWomen)
```

```
    {
        SceneManager.LoadScene("bucWomen");
    }
    public void womenSouthButton(string womenSouth)
    {
        SceneManager.LoadScene("womenSouth");
    }
    public void womensNorthButton(string womensNorth)
    {
        SceneManager.LoadScene("womensNorth");
    }
    public void littlejohnButton(string littlejohn)
    {
        SceneManager.LoadScene("littlejohn");
    }
    public void bucTennisButton(string bucTennis)
    {
        SceneManager.LoadScene("bucTennis");
    }
/*
    public void tennisButton(string tennis)
    {
        SceneManager.LoadScene("tennis");
    }
*/
    public void bucAlumniButton(string bucAlumni)
    {
        SceneManager.LoadScene("bucAlumni");
    }
/*
    public void alumniBaseballButton(string alumniBaseball)
    {
        SceneManager.LoadScene("alumniBaseball");
    }

    public void baseballButton(string baseball)
    {
        SceneManager.LoadScene("baseball");
    }
*/
    public void alumniBrewerButton(string alumniBrewer)
    {
        SceneManager.LoadScene("alumniBrewer");
    }
    public void russellResidenceButton(string russellResidence)
    {
        SceneManager.LoadScene("russellResidence");
    }
    public void russellWestButton(string russellWest)
    {
        SceneManager.LoadScene("russellWest");
    }
    public void learningCenterButton(string learningCenter)
    {
        SceneManager.LoadScene("learningCenter");
    }
    public void alumniPoolButton(string alumniPool)
    {
        SceneManager.LoadScene("alumniPool");
    }
    public void poolButton(string pool)
    {
        SceneManager.LoadScene("pool");
    }
/*
    public void comCenterButton(string comCenter)
```

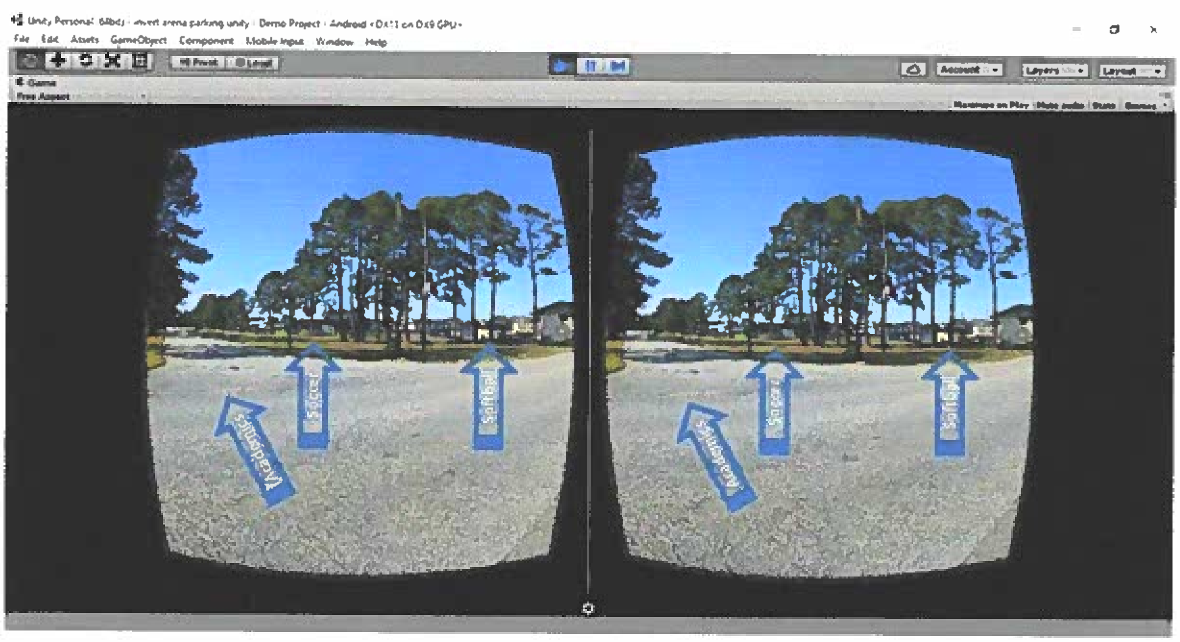
```
    {
        SceneManager.LoadScene("comCenter");
    }

    public void parkButton(string park)
    {
        SceneManager.LoadScene("park");
    }
*/
    public void alumniWhittingtonButton(string alumniWhittington)
    {
        SceneManager.LoadScene("alumniWhittington");
    }
    public void alumniQuadButton(string alumniQuad)
    {
        SceneManager.LoadScene("alumniQuad");
    }
    public void quadsButton(string quads)
    {
        SceneManager.LoadScene("quads");
    }
    public void wellnessAlumniButton(string wellnessAlumni)
    {
        SceneManager.LoadScene("wellnessAlumni");
    }
    public void physicalPlantButton(string physicalPlant)
    {
        SceneManager.LoadScene("physicalPlant");
    }
    public void artButton(string art)
    {
        SceneManager.LoadScene("art");
    }
    public void excellenceWellnessButton(string excellenceWellness)
    {
        SceneManager.LoadScene("excellenceWellness");
    }
}
```

```
// http://www.andrewnoske.com/wiki/Unity_-_Panorama_Viewer
//
using UnityEngine;
using UnityEditor;

public class InvertedSphere : EditorWindow
{
    private string st = "1.0";

    [MenuItem("GameObject/Create Other/Inverted Sphere...")]
    public static void ShowWindow()
    {
        EditorWindow.GetWindow(typeof(InvertedSphere));
    }

    public void OnGUI()
    {
        GUILayout.Label("Enter sphere size:");
        st = GUILayout.TextField(st);

        float f;
        if (!float.TryParse(st, out f))
            f = 1.0f;
        if (GUILayout.Button("Create Inverted Sphere"))
        {
            CreateInvertedSphere(f);
        }
    }

    private void CreateInvertedSphere(float size)
    {
        GameObject go = GameObject.CreatePrimitive(PrimitiveType.Sphere);
        MeshFilter mf = go.GetComponent<MeshFilter>();
        Mesh mesh = mf.sharedMesh;

        GameObject goNew = new GameObject();
        goNew.name = "Inverted Sphere";
        MeshFilter mfNew = goNew.AddComponent<MeshFilter>();
        mfNew.sharedMesh = new Mesh();


        //Scale the vertices;
        Vector3[] vertices = mesh.vertices;
        for (int i = 0; i < vertices.Length; i++)
            vertices[i] = vertices[i] * size;
        mfNew.sharedMesh.vertices = vertices;

        // Reverse the triangles
        int[] triangles = mesh.triangles;
        for (int i = 0; i < triangles.Length; i += 3)
        {
            int t = triangles[i];
            triangles[i] = triangles[i + 2];
            triangles[i + 2] = t;
        }
        mfNew.sharedMesh.triangles = triangles;

        // Reverse the normals;
        Vector3[] normals = mesh.normals;
        for (int i = 0; i < normals.Length; i++)
            normals[i] = -normals[i];
        mfNew.sharedMesh.normals = normals;


        mfNew.sharedMesh.uv = mesh.uv;
        mfNew.sharedMesh.uv2 = mesh.uv2;
```

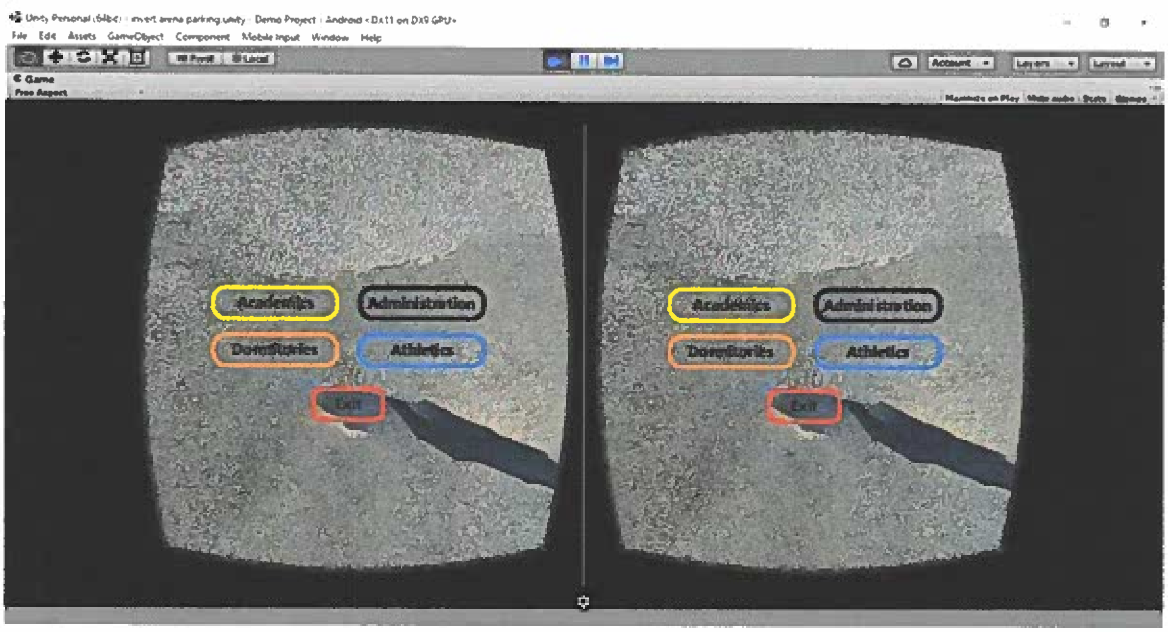
```
        mfNew.sharedMesh.RecalculateBounds();

        // Add the same material that the original sphere used
        MeshRenderer mr = goNew.AddComponent<MeshRenderer>();
        mr.sharedMaterial = go.GetComponent<Renderer>().sharedMaterial;

        DestroyImmediate(go);
    }
}
```

Button Prototypes:

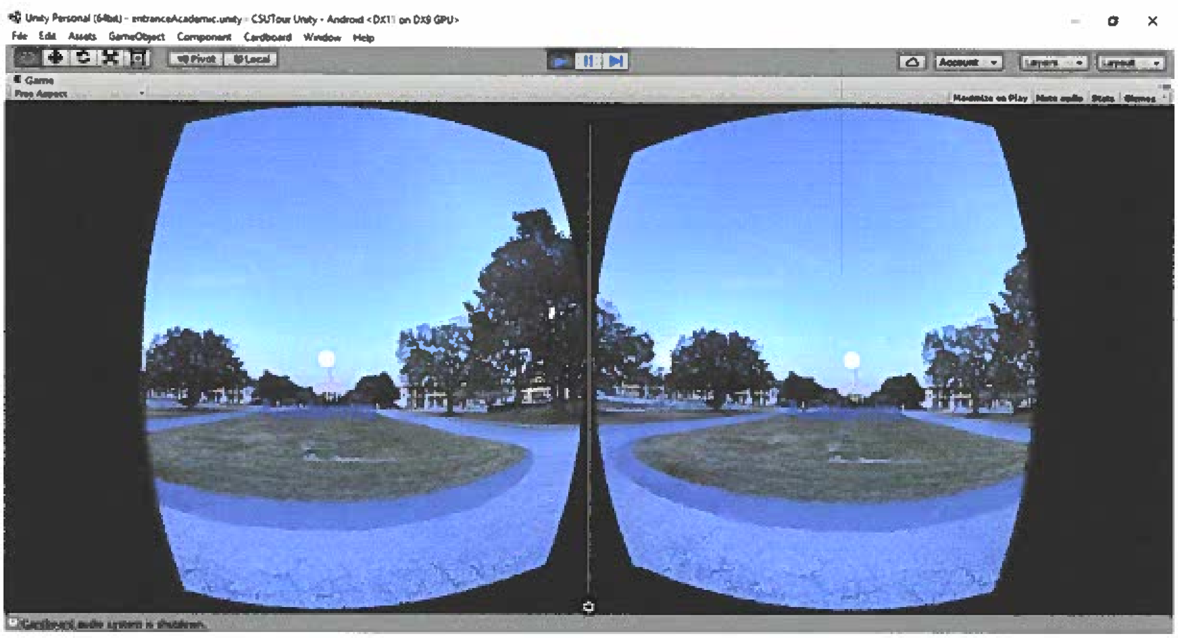
This app was created using Unity 3D with Google Cardboard SDK. The photospheres were captured using the Google Street App for I-Phones. Below are my initial ideas on creating buttons that would allow the using to navigate through the app. Each sphere would contain directional arrows to allow the user to navigate to the next adjacent scene they wish to view. The first two pictures depict how arrows would be used to show the direction the user would be moving.



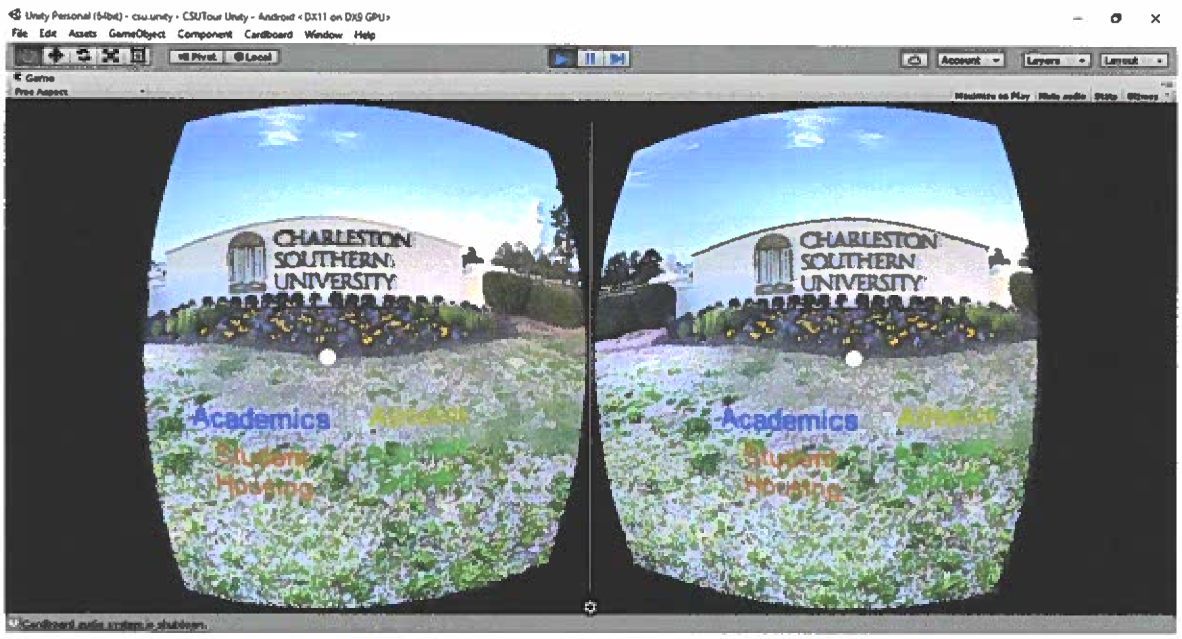
In addition to the arrows about, I wanted to create a way for the user to jump to various areas of the campus. The initial prototype for these buttons are shown below. These buttons would be locates at the center of the bottom, looking straight down, and would have a menu consisting of Academics, Administration, Dormitories, and Athletics. Below these buttons would be an exit button that would allow the user to close the app.



After learning more about user interfaces in Human Computer Interaction, I elected to us the text shown below for navigation in this app. I also elected to remove the full menu from the bottom of the scene and only provide a link back to the opening scene. I also learned that the exit button I had hoped to use would not work because the phone closes the app, not the app.



On the opening scene, pictured below, I chose to stay uniform with the buttons throughout the app and only used the names as buttons. I kept the color coding consistent throughout the app to help the user know the type of area the link would take them to.



Obstacles Encountered:

1. The quality of the photospheres obtained using a Samsung Galaxy 5 with the Google Camera App.
    a. Some of photospheres were incomplete.
    b. Some of the photospheres had frames that were not properly aligning.
    c. To Solve these problems, I used an I-Phone 6 with the Google Street app
2. Learning how to develop an app using Unity 3D
    a. To overcome this obstacle, I created the demo app to familiarize myself with the platform.
    b. After this, I used the Unity 3D website to help with any other issues
3. Positioning of the camera to view the photosphere
    a. Learned I need to position the camera on the inside of the photosphere.
    b. Once inside, I realized I needed to invert the photo to view to correct side and reverse the photo.
    c. This was accomplished by using an inverted sphere and changing the orientation to -1

4. Creating and positioning of the Navigation buttons.
   a. The buttons used in Unity to not fit into the environment I was trying to create
   b. To overcome this I:
      i. Used the Mesh Text to create text blocks
      ii. Used the Box Collider to change the text blocks into buttons
      iii. Used the Cardboard Reticle provided in to SDK to select the button
   c. To position the buttons, I placed them on the screen to set the initial position, then adjusted the position on the x, y, and z axis
5. Correcting the horizontal view on the Android Phone
   a. I learned that if you opened the app with the phone in the vertical position, when you placed the phone inside Google Cardboard, the screen would not properly resize.
   b. To correct this, I changed the display orientation to only use left horizontal.

Resources:

https://developers.google.com/vr/?hl=en

https://unity3d.com/

https://www.sitepoint.com/building-a-google-cardboard-vr-app-in-unity/

http://www.andrewnoske.com/wiki/Unity_-_Panorama_Viewer

http://www.pepwuper.com/portfolio/google-cardboard-vr-with-unity-navigation-pathfinding/

https://play.google.com/store/apps/details?id=com.smartinfolab.vr3dcamera&hl=en

https://www.google.com/design/spec-vr/interactive-patterns/setup.html#

http://makezine.com/projects/getting-started-with-virtual-reality-building-for-google-cardboard/

http://bernieroehl.com/360stereoinunity/

https://play.google.com/store/apps/details?id=kamil.application.mainApp.freesimplescreencast